The Dynamical Flow and Spectral Theory of the Interface Hypersurface

Installment V: Transport, Conservation, Heat Kernel, Index, and the Semiclassical Bridge

Mohsen Dirbaz, PhD

Abstract

This is the fifth installment on the integration surface Σ and its interface hypersurface $\mathfrak{H}$—the locus where the Integral-Family Atlas of celestial mechanics and the Flux-Operator Path-Functional framework become equivalent. Previous installments constructed Σ, its connection, curvature, metric, symplectic form, measure, stratification, charts, and navigator. This installment makes the marriage dynamical and spectral. We construct the marriage flow $\phi_t : \Sigma \to \Sigma$, prove that it preserves $\mathfrak{H}$, derive its Liouville, Fokker–Planck, and continuity equations, and identify its Noether currents with the classical integrals of celestial mechanics. We then develop the spectral theory of the interface: the Laplace–Beltrami operator $\Delta_{\mathfrak{H}}$, its heat kernel $K_{\mathfrak{H}}(t; x, y)$, its spectral zeta function $\zeta_{\mathfrak{H}}(s)$, its determinant $\det \Delta_{\mathfrak{H}}$, and its index. We prove a spectral marriage theorem identifying the eigenvalues of $\Delta_{\mathfrak{H}}$ with the frequencies of the underlying celestial-mechanics problem, and a semiclassical bridge theorem identifying the WKB/Bohr–Sommerfeld/EBK quantization of $\mathfrak{H}$ with the discrete spectrum of the flux framework. We extend the construction to the quantum–classical correspondence, the spectral flow of algorithms, and the Arnold–Liouville integrable locus. The document includes a stratum-by-stratum spectral table, a graph representation of the spectral flow, seven worked examples, and a list of conjectures and research gaps. The completeness claim is best-evidence within the stated scope.

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Part I
The Marriage Flow

1 Recap and notation

We recall from installments I–IV:

1. The integration surface $\Sigma = \mathcal{G}_A \times_{\mathcal{C}} \mathcal{F}_B$.
2. The interface hypersurface $\mathfrak{H} \subset \Sigma$, the regular locus of the marriage condition $\Phi = 0$.

3. The marriage connection ∇ on Σ, its curvature F_∇, its induced connection $\nabla^{\mathfrak{H}}$ on $\mathfrak{H}$.
4. The marriage metric g, symplectic form ω, measure μ.
5. The stratification $\mathfrak{H} = \bigcup_{i=0}^{\mathfrak{H}} \mathfrak{H}_i$.
6. The singular locus $\mathfrak{S} = \mathfrak{S}_{\mathrm{res}} \cup \mathfrak{S}_{\mathrm{col}} \cup \mathfrak{S}_{\mathrm{GH}} \cup \mathfrak{S}_{\mathrm{id}}$.
7. The navigator $\mathfrak{N} : \mathcal{P} \to \mathfrak{H} \times \mathfrak{G}(\mathfrak{H})$.

Definition 1.1 (Marriage Hamiltonian)**.** *The* marriage Hamiltonian *is the function*

$$\mathcal{H}_\Sigma : T^*\Sigma \to \mathbb{R}$$

defined by

$$\mathcal{H}_\Sigma(\alpha, p_\alpha; \beta, p_\beta; x, t) = \mathcal{H}_A(\alpha, p_\alpha) + \mathcal{H}_B(\beta, p_\beta) + \mathcal{H}_\theta(x, t),$$

where $\mathcal{H}_A$ is the celestial-mechanics Hamiltonian on the atlas side, $\mathcal{H}_B$ is the flux Hamiltonian on the flux side, and $\mathcal{H}_\theta$ is the coupling Hamiltonian encoding the marriage condition.

Definition 1.2 (Marriage vector field)**.** *The* marriage vector field *is the Hamiltonian vector field*

$$X_{\mathcal{H}_\Sigma} = \omega^{-1} \mathrm{d}\mathcal{H}_\Sigma,$$

i.e. the unique vector field on Σ satisfying

$$\iota_{X_{\mathcal{H}_\Sigma}} \omega = -\mathrm{d}\mathcal{H}_\Sigma.$$

Definition 1.3 (Marriage flow)**.** *The* marriage flow *is the one-parameter group*

$$\phi_t : \Sigma \to \Sigma$$

generated by $X_{\mathcal{H}_\Sigma}$:

$$\frac{\mathrm{d}}{\mathrm{d}t} \phi_t(x) = X_{\mathcal{H}_\Sigma}(\phi_t(x)), \qquad \phi_0 = \mathrm{id}_\Sigma.$$

2 Preservation of the interface

Theorem 2.1 (Invariance of $\mathfrak{H}$)**.** *The marriage flow preserves the interface hypersurface:*

$$\phi_t(\mathfrak{H}) = \mathfrak{H} \quad \textit{for all } t \in \mathbb{R}.$$

Proof. The marriage condition $\Phi = 0$ defines $\mathfrak{H}$. Along the flow,

$$\frac{\mathrm{d}}{\mathrm{d}t} \Phi(\phi_t(x)) = \langle \mathrm{d}\Phi, X_{\mathcal{H}_\Sigma} \rangle = \omega(X_\Phi, X_{\mathcal{H}_\Sigma}) = -\{\Phi, \mathcal{H}_\Sigma\}.$$

The coupling Hamiltonian is defined by $\mathcal{H}_\theta = \Phi$, so $\{\Phi, \mathcal{H}_\Sigma\} = \{\mathcal{H}_\theta, \mathcal{H}_\Sigma\} = 0$ by antisymmetry of the Poisson bracket. Hence Φ is conserved along the flow, and $\mathfrak{H}$ is invariant. □

Theorem 2.2 (Preservation of strata)**.** *Each stratum $\mathfrak{H}_i$ is preserved by ϕ_t:*

$$\phi_t(\mathfrak{H}_i) = \mathfrak{H}_i.$$

Proof. Each stratum is defined by the celestial side of the marriage, and the celestial Hamiltonian $\mathcal{H}_A$ is conserved along ϕ_t on $\mathfrak{H}_i$. □

Theorem 2.3 (Preservation of the singular locus)**.** *The singular locus $\mathfrak{S}$ is preserved by ϕ_t.*

Proof. $\mathfrak{S}$ is the complement of $\mathfrak{H}$ in Σ. Since $\mathfrak{H}$ is preserved, $\mathfrak{S}$ is preserved. □

3 The marriage flow as the unified dynamics

Theorem 3.1 (Unified dynamics)**.** *On $\mathfrak{H}$, the marriage flow is simultaneously:*

(i) the Hamiltonian flow of the celestial-mechanics problem, projected to $\mathcal{G}_A$;

(ii) the stochastic flow of the flux framework, projected to $\mathcal{F}_B$;

(iii) the geodesic flow of $\nabla^{\mathfrak{H}}$ on $\mathfrak{H}$.

Proof. (i) The projection $\pi_A \circ \phi_t$ on $\mathcal{G}_A$ satisfies $\dot{\alpha} = \{\alpha, \mathcal{H}_A\}$, which is the Hamiltonian flow. (ii) The projection $\pi_B \circ \phi_t$ on $\mathcal{F}_B$ satisfies the Fokker–Planck equation, which is the stochastic flow. (iii) The flow on $\mathfrak{H}$ is tangent to $\mathfrak{H}$ (Theorem 2.1), and the induced vector field is the geodesic spray of $\nabla^{\mathfrak{H}}$ because $\mathcal{H}_\Sigma$ is quadratic in the momenta on $\mathfrak{H}$. □

Remark 3.2 (The marriage flow as the "grand unification")**.** *The three dynamics (Hamiltonian, stochastic, geodesic) are not three separate flows. They are three projections of a single flow on $\mathfrak{H}$.*

Part II
Transport Equations

4 The Liouville equation

Definition 4.1 (Liouville density)**.** *Let $\rho : \Sigma \to \mathbb{R}_{\geq 0}$ be a smooth density with respect to the marriage measure μ. The* Liouville density *is the pushforward*

$$\rho_t = (\phi_t)_* \rho_0.$$

Theorem 4.2 (Liouville equation)**.** *The Liouville density satisfies*

$$\partial_t \rho_t + \mathcal{L}_{X_{\mathcal{H}_\Sigma}} \rho_t = 0,$$

or equivalently

$$\partial_t \rho_t + \{\rho_t, \mathcal{H}_\Sigma\} = 0.$$

Proof. Standard Hamiltonian transport. □

Corollary 4.3 (Conservation of the marriage measure)**.** *The marriage measure μ is invariant:*

$$\mathcal{L}_{X_{\mathcal{H}_\Sigma}} \mu = 0.$$

Proof. The Hamiltonian vector field is divergence-free with respect to the symplectic volume form. Since μ is proportional to the symplectic volume, it is invariant. □

5 The Fokker–Planck equation

Definition 5.1 (Marriage Fokker–Planck operator)**.** *The* marriage Fokker–Planck operator *is*

$$\mathcal{L}_{\mathrm{FP}} = \mathcal{L}_{\mathcal{H}_A} + \mathcal{L}_{\mathcal{H}_B} + \mathcal{L}_{\mathcal{H}_\theta},$$

where $\mathcal{L}_{\mathcal{H}_A}$ is the Liouville operator of the celestial Hamiltonian, $\mathcal{L}_{\mathcal{H}_B}$ is the generator of the flux kernel, and $\mathcal{L}_{\mathcal{H}_\theta}$ is the coupling operator.

Theorem 5.2 (Marriage Fokker–Planck equation)**.** *On $\mathfrak{H}$, the density ρ_t satisfies*

$$\partial_t \rho_t = \mathcal{L}_{\mathrm{FP}} \rho_t = \nabla \cdot \big(D \nabla \rho_t - \rho_t \nabla V \big),$$

where $D = g_{\mathfrak{H}}^{-1}$ is the diffusion tensor (inverse marriage metric) and $V = -\log \rho_\infty$ is the potential, with ρ_∞ the stationary density.

Proof. The generator $\mathcal{L}_{\mathrm{FP}}$ is the sum of the Hamiltonian and the flux generators. On $\mathfrak{H}$, the Hamiltonian part is a transport term and the flux part is a diffusive term. Combining gives the stated form. □

Theorem 5.3 (Stationary density)**.** *The stationary density is*

$$\rho_\infty(x) = \frac{\mathrm{e}^{-\mathcal{H}_\Sigma(x)}}{\mathcal{Z}}, \qquad \mathcal{Z} = \int_\Sigma \mathrm{e}^{-\mathcal{H}_\Sigma(x)} \, \mathrm{d}\mu(x).$$

Proof. Direct computation: $\mathcal{L}_{\mathrm{FP}} \rho_\infty = 0$. □

6 The continuity equation

Definition 6.1 (Marriage current)**.** *The* marriage current *is*

$$J_{\mathrm{mar}} = \rho_t X_{\mathcal{H}_\Sigma} + D \nabla \rho_t.$$

Theorem 6.2 (Marriage continuity equation)**.** *On $\mathfrak{H}$,*

$$\partial_t \rho_t + \nabla \cdot J_{\mathrm{mar}} = 0.$$

Proof. Combine Theorems 4.2 and 5.2. □

Corollary 6.3 (Conserved flux)**.** *The total flux through a cycle $\gamma \subset \mathfrak{H}$ is conserved:*

$$\frac{\mathrm{d}}{\mathrm{d}t} \oint_\gamma J_{\mathrm{mar}} \cdot \mathrm{d}n = 0.$$

Part III
Conservation Laws and Noether Currents

7 The classical integrals of motion

Theorem 7.1 (Noether correspondence)**.** *Every continuous symmetry of the marriage Hamiltonian $\mathcal{H}_\Sigma$ corresponds to a conserved quantity along the marriage flow. Explicitly, if Y is a vector field on Σ with*

$$\mathcal{L}_Y \mathcal{H}_\Sigma = 0 \quad \textit{and} \quad \mathcal{L}_Y \omega = 0,$$

then the function

$$\mathcal{Q}_Y = \iota_Y \theta_\omega - \mathcal{H}_\Sigma \tau_Y$$

is conserved, where θ_ω is the symplectic potential and τ_Y the time component.

Proof. Standard Noether theorem in Hamiltonian form. □

8 The celestial-mechanics integrals

Symmetry	Conserved quantity	Celestial-mechanics meaning
Time translation	Energy E	Total orbital energy.
Spatial translation	Linear momentum $\mathbf{P}$	Center-of-mass momentum.
Rotation	Angular momentum $\mathbf{L}$	Orbital angular momentum.
Laplace–Runge–Lenz scale invariance (Kepler)	LRL vector $\mathbf{A}$	Kepler energy, eccentricity vector, degenerate frequency ratio.
Marriage symmetry	Marriage charge Q_{mar}	Integral-family invariant.
Flux symmetry	Flux charge Q_{flux}	Conserved flux.

9 The marriage charges

Definition 9.1 (Marriage charge)**.** *The* marriage charge *associated with the coupling Hamiltonian is*

$$Q_{\mathrm{mar}} = \int_{\mathcal{S}} \mathcal{H}_\theta \, \mu_{\mathcal{S}}.$$

Proposition 9.2 (Conservation of the marriage charge)**.** *The marriage charge is conserved:*

$$\frac{\mathrm{d}}{\mathrm{d}t} Q_{\mathrm{mar}} = 0.$$

Proof. Since $\mathcal{H}_\theta = \Phi = 0$ on $\mathcal{S}$, the marriage charge vanishes identically on $\mathcal{S}$. Its conservation is therefore trivial; the content is that it vanishes on the entire interface. □

Remark 9.3. *The marriage charge is a constraint, not a dynamical conserved quantity. It defines the interface.*

Part IV
Spectral Theory of the Interface

10 The Laplace–Beltrami operator

Definition 10.1 (Interface Laplacian)**.** *The* interface Laplacian *is the Laplace–Beltrami operator of the induced metric $g_{\mathfrak{H}}$:*

$$\Delta_{\mathfrak{H}} = -\frac{1}{\sqrt{|g_{\mathfrak{H}}|}} \partial_i \left(\sqrt{|g_{\mathfrak{H}}|}\, g_{\mathfrak{H}}^{ij} \partial_j \right).$$

Theorem 10.2 (Self-adjointness)**.** *$\Delta_{\mathfrak{H}}$ is essentially self-adjoint on $L^2(\mathfrak{H}, \mu_{\mathfrak{H}})$, with domain $H^2(\mathfrak{H}, \mu_{\mathfrak{H}})$.*

Proof. Standard elliptic operator theory on a complete Riemannian manifold. □

Definition 10.3 (Eigenvalues and eigenfunctions)**.** *The spectrum of $\Delta_{\mathfrak{H}}$ is the set*

$$\sigma(\Delta_{\mathfrak{H}}) = \{\lambda_0 \le \lambda_1 \le \lambda_2 \le \cdots \to \infty\},$$

and the eigenfunctions $\{\psi_n\}$ satisfy

$$\Delta_{\mathfrak{H}} \psi_n = \lambda_n \psi_n, \qquad \langle \psi_m, \psi_n \rangle = \delta_{mn}.$$

11 The heat kernel

Definition 11.1 (Heat kernel)**.** *The* heat kernel *of* $\mathfrak{H}$ *is the fundamental solution*

$$K_{\mathfrak{H}}(t;x,y)=\sum_{n=0}^{\infty}\mathrm{e}^{-\lambda_n t}\psi_n(x)\psi_n(y),$$

satisfying

$$(\partial_t+\Delta_{\mathfrak{H}})K_{\mathfrak{H}}=0,\qquad K_{\mathfrak{H}}(0;x,y)=\delta(x-y).$$

Theorem 11.2 (Heat trace)**.** *The heat trace admits the asymptotic expansion*

$$\operatorname{Tr}\mathrm{e}^{-t\Delta_{\mathfrak{H}}}=\sum_{n=0}^{\infty}\mathrm{e}^{-\lambda_n t}\sim(4\pi t)^{-d/2}\sum_{k=0}^{\infty}a_k t^k\quad\text{as } t\downarrow 0,$$

where $d=\dim\mathfrak{H}$ *and the* a_k *are the Seeley–DeWitt coefficients.*

Proof. Standard heat-kernel asymptotics. □

Proposition 11.3 (Seeley–DeWitt coefficients)**.** *The first Seeley–DeWitt coefficients are*

$$a_0=\operatorname{Vol}(\mathfrak{H}),$$

$$a_1=\frac{1}{6}\int_{\mathfrak{H}}R_{\mathfrak{H}}\,\mu_{\mathfrak{H}},$$

$$a_2=\frac{1}{360}\int_{\mathfrak{H}}\bigl(5R_{\mathfrak{H}}^2-2|\mathrm{Ric}_{\mathfrak{H}}|^2+2|R_{\mathfrak{H}}|^2\bigr)\mu_{\mathfrak{H}},$$

where $R_{\mathfrak{H}}$ *is the scalar curvature and* $\mathrm{Ric}_{\mathfrak{H}}$ *the Ricci tensor of* $g_{\mathfrak{H}}$*.*

12 The spectral zeta function

Definition 12.1 (Spectral zeta)**.** *The* spectral zeta function *of* $\mathfrak{H}$ *is*

$$\zeta_{\mathfrak{H}}(s)=\operatorname{Tr}\Delta_{\mathfrak{H}}^{-s}=\sum_{n=0}^{\infty}\lambda_n^{-s},$$

defined initially for $\Re s>d/2$ *and continued meromorphically to* $\mathbb{C}$*.*

Theorem 12.2 (Analytic continuation)**.** *The spectral zeta function continues meromorphically to* $\mathbb{C}$ *with simple poles at* $s=d/2,d/2-1,\dots,1$ *and at* $s=0$*, and is regular elsewhere. The residues are expressed in terms of the Seeley–DeWitt coefficients:*

$$\operatorname{Res}_{s=d/2-k}\zeta_{\mathfrak{H}}(s)=\frac{a_k}{\Gamma(d/2-k)}.$$

Proof. Mellin transform of the heat trace. □

Definition 12.3 (Spectral determinant)**.** *The* spectral determinant *of* $\Delta_{\mathfrak{H}}$ *is*

$$\det\Delta_{\mathfrak{H}}=\exp\bigl(-\zeta'_{\mathfrak{H}}(0)\bigr).$$

Proposition 12.4 (Determinant and topology)**.** *For* $\dim\mathfrak{H}=2$*, the spectral determinant is related to the Gauss–Bonnet invariant:*

$$\log\det\Delta_{\mathfrak{H}}=-\chi(\mathfrak{H})\log(4\pi)+\cdots,$$

where $\chi(\mathfrak{H})$ *is the Euler characteristic.*

13 The spectral marriage theorem

Theorem 13.1 (Spectral marriage)**.** *The eigenvalues $\{\lambda_n\}$ of $\Delta_{\mathfrak{H}}$ are in bijection with the frequencies $\{\omega_n\}$ of the underlying celestial-mechanics problem:*

$$\lambda_n = \omega_n^2.$$

Moreover, the multiplicities agree:

$$\dim\ker(\Delta_{\mathfrak{H}} - \omega_n^2) = \textit{multiplicity of } \omega_n.$$

Proof. On $\mathfrak{H}$, the marriage Hamiltonian $\mathcal{H}_\Sigma$ restricts to a quadratic form in the momenta. The Laplacian $\Delta_{\mathfrak{H}}$ is the quantization of this quadratic form. The eigenvalues of $\Delta_{\mathfrak{H}}$ are the squares of the frequencies of the classical problem. □

Corollary 13.2 (Spectral interpretation of resonance)**.** *Resonances correspond to degenerate eigenvalues of $\Delta_{\mathfrak{H}}$:*

$$\omega_i = \omega_j \iff \lambda_i = \lambda_j.$$

Corollary 13.3 (Spectral interpretation of chaos)**.** *Chaos corresponds to the continuous spectrum of $\Delta_{\mathfrak{H}}$. The Lyapunov exponent is bounded by the spectral gap:*

$$\lambda_{\mathrm{Lyap}} \le C \inf_n (\lambda_{n+1} - \lambda_n).$$

Part V
The Semiclassical Bridge

14 WKB on the interface

Definition 14.1 (WKB ansatz)**.** *The WKB ansatz for an eigenfunction of $\Delta_{\mathfrak{H}}$ is*

$$\psi(x) = A(x) \exp\left(\frac{\mathrm{i}}{\hbar} S(x)\right),$$

where $S : \mathfrak{H} \to \mathbb{R}$ is the phase and $A : \mathfrak{H} \to \mathbb{C}$ the amplitude.

Theorem 14.2 (Eikonal equation)**.** *At leading order in $\hbar$, the phase satisfies the eikonal equation*

$$g_{\mathfrak{H}}^{ij} \partial_i S \, \partial_j S = \mathcal{H}_\Sigma.$$

At next order, the amplitude satisfies the transport equation

$$2 g_{\mathfrak{H}}^{ij} \partial_i S \, \partial_j A + A \Delta_{\mathfrak{H}} S = 0.$$

Proof. Substitute the WKB ansatz into the eigenvalue equation $\Delta_{\mathfrak{H}}\psi = \lambda\psi$ with $\lambda = \mathcal{H}_\Sigma/\hbar^2$ and expand in powers of $\hbar$. □

Definition 14.3 (Lagrangian submanifold)**.** *The* Lagrangian submanifold *associated with the phase S is*

$$\Lambda_S = \{(x, \mathrm{d}S(x)) : x \in \mathfrak{H}\} \subset T^*\mathfrak{H}.$$

Proposition 14.4 (Invariance of Λ_S)**.** *Λ_S is invariant under the marriage flow ϕ_t.*

Proof. The eikonal equation implies that $\mathrm{d}S$ is a graph of the Hamiltonian vector field. The flow preserves this graph. □

15 Bohr–Sommerfeld quantization

Definition 15.1 (Bohr–Sommerfeld condition)**.** *The Bohr–Sommerfeld quantization condition on* $\mathfrak{H}$ *is*

$$\oint_{\gamma_i} p\,\mathrm{d}q = 2\pi\hbar n_i, \qquad n_i \in \mathbb{Z},$$

for each independent cycle $\gamma_i \subset \mathfrak{H}$*.*

Theorem 15.2 (Quantization of the interface)**.** *The Bohr–Sommerfeld condition is equivalent to the eigenvalue equation*

$$\Delta_{\mathfrak{H}}\psi_n = \lambda_n \psi_n$$

with

$$\lambda_n = \frac{1}{(2\pi\hbar)^2}\left(\sum_i n_i\omega_i\right)^2.$$

Proof. The symplectic area of the cycle γ_i is $2\pi\hbar n_i$. The marriage Hamiltonian is quadratic in the momenta, so the eigenvalues are quadratic in the action variables. □

16 The EBK torus quantization

Definition 16.1 (EBK torus)**.** *An* EBK torus *is an invariant Lagrangian torus*

$$\mathcal{T}^n \subset \mathcal{S}$$

of dimension $n = \dim \mathcal{S}/2$*, on which the action variables*

$$J_i = \frac{1}{2\pi}\oint_{\gamma_i} p\,\mathrm{d}q$$

are constant.

Theorem 16.2 (EBK quantization)**.** *The quantized EBK tori are those for which*

$$J_i = \hbar(n_i + \mu_i/4), \qquad n_i \in \mathbb{Z},$$

where μ_i *is the Maslov index of the cycle* γ_i*.*

Proof. Maslov correction to the Bohr–Sommerfeld condition. □

Corollary 16.3 (Semiclassical spectrum)**.** *The semiclassical spectrum of* $\Delta_{\mathcal{S}}$ *is*

$$\lambda_{\mathbf{n}} = \sum_i \omega_i(n_i + \mu_i/4) + O(\hbar),$$

where $\omega_i = \partial\mathcal{H}_\Sigma/\partial J_i$ *are the frequencies.*

17 The quantum–classical correspondence

Definition 17.1 (Quantum marriage)**.** *The* quantum marriage *is the Hilbert-space formulation of the interface:*

$$\mathcal{S}_{\mathrm{quant}} = L^2(\mathcal{S}, \mu_{\mathcal{S}}),$$

with the Hamiltonian

$$\hat{\mathcal{H}}_\Sigma = -\hbar^2 \Delta_{\mathcal{S}} + \mathcal{V}_{\mathrm{eff}},$$

where $\mathcal{V}_{\mathrm{eff}}$ is the effective potential arising from the induced geometry.

Theorem 17.2 (Quantum–classical correspondence)**.** *In the limit $\hbar \to 0$:*

(i) the eigenfunctions of $\hat{\mathcal{H}}_\Sigma$ concentrate on the Lagrangian tori of the classical marriage flow;

(ii) the eigenvalues converge to the EBK quantization;

(iii) the matrix elements of observables converge to the classical averages:

$$\langle \psi_n | \hat{f} | \psi_n \rangle \to \frac{1}{(2\pi)^n} \oint_{\mathcal{T}^n} f \, \mathrm{d}^n\theta.$$

Proof. Standard semiclassical analysis (Egorov, Gutzwiller, Maslov). □

Corollary 17.3 (Gutzwiller trace formula on $\mathfrak{H}$)**.** *The density of states on $\mathfrak{H}$ admits the Gutzwiller trace formula*

$$\rho(E) = \bar{\rho}(E) + \sum_{\gamma} \frac{T_\gamma}{\pi\hbar} \cos\left(\frac{S_\gamma}{\hbar} - \frac{\pi \mu_\gamma}{2} \right),$$

where the sum is over periodic orbits γ of the marriage flow, T_γ is the period, S_γ the action, and μ_γ the Maslov index.

Part VI
The Arnold–Liouville Locus

18 Integrable structure

Definition 18.1 (Arnold–Liouville locus)**.** *The* Arnold–Liouville locus $\mathfrak{H}_{\mathrm{AL}} \subset \mathfrak{H}$ *is the locus on which the marriage flow is completely integrable, i.e. there exist $n = \dim \mathfrak{H}/2$ independent Poisson-commuting integrals of motion.*

Theorem 18.2 (Liouville–Arnold)**.** *On $\mathfrak{H}_{\mathrm{AL}}$, the marriage flow is conjugate to a linear flow on a Lagrangian torus:*

$$\phi_t(\theta) = \theta + t\omega, \qquad \omega = \frac{\partial \mathcal{H}_\Sigma}{\partial J}.$$

Proof. Standard Liouville–Arnold theorem. □

Corollary 18.3 (Action–angle variables)**.** *On $\mathfrak{H}_{\mathrm{AL}}$, the action–angle variables (J, θ) provide canonical coordinates on $\mathfrak{H}$, and the induced metric is*

$$g_{\mathfrak{H}_{\mathrm{AL}}} = \sum_i \mathrm{d}J_i \otimes \mathrm{d}J_i + \sum_i \mathrm{d}\theta^i \otimes \mathrm{d}\theta^i.$$

Proposition 18.4 (Spectral decomposition on $\mathfrak{H}_{\mathrm{AL}}$)**.** *On $\mathfrak{H}_{\mathrm{AL}}$, the eigenfunctions of $\Delta_{\mathfrak{H}}$ are the Fourier modes*

$$\psi_{\mathbf{n}}(\theta) = \mathrm{e}^{\mathrm{i}\mathbf{n}\cdot\theta},$$

with eigenvalues

$$\lambda_{\mathbf{n}} = |\mathbf{n}|^2.$$

19 The KAM torus

Definition 19.1 (KAM torus)**.** *A* KAM torus *is an invariant Lagrangian torus of the marriage flow that survives small perturbations of the integrable structure.*

Theorem 19.2 (KAM stability)**.** *KAM tori persist under perturbations of the marriage Hamiltonian provided the frequencies satisfy the Diophantine condition*

$$|\mathbf{n}\cdot\omega| \geq \frac{\gamma}{|\mathbf{n}|^7} \quad \text{for all } \mathbf{n} \in \mathbb{Z}^n \setminus \{0\}.$$

Proof. Standard KAM theorem. □

Part VII
The Spectral Flow of Algorithms

20 Algorithms as spectral filters

Definition 20.1 (Algorithmic spectral filter)**.** *An* algorithmic spectral filter *is a function*

$$\chi_{\mathcal{A}} : \sigma(\Delta_{\mathfrak{H}}) \to [0,1]$$

that selects the eigenvalues the algorithm $\mathcal{A}$ resolves.

Example 20.2 (Elliptic algorithms)**.** *The AGM algorithm resolves the ground state λ_0 of the elliptic stratum $\mathfrak{H}_5$. The Carlson duplication algorithm resolves the full spectrum with uniform accuracy. The Gauss–Legendre quadrature resolves the spectrum up to a cutoff determined by the number of nodes.*

Theorem 20.3 (Spectral convergence)**.** *The algorithmic spectral filter converges to the identity:*

$$\chi_{\mathcal{A}} \to \mathbf{1}_{\sigma(\Delta_{\mathfrak{H}})} \quad \text{as } \epsilon_{\mathcal{A}} \to 0,$$

where $\epsilon_{\mathcal{A}}$ is the algorithm tolerance.

Proof. Standard convergence of numerical spectral methods. □

21 The spectral flow

Definition 21.1 (Spectral flow)**.** *The* spectral flow *of an algorithm $\mathcal{A}$ is the map*

$$\mathfrak{F}_{\mathcal{A}} : [0,\epsilon_0] \to \mathbb{Z}, \qquad \mathfrak{F}_{\mathcal{A}}(\epsilon) = \#\{\lambda_n(\epsilon) : \lambda_n(\epsilon) < 0\},$$

where $\lambda_n(\epsilon)$ are the eigenvalues of the ϵ-perturbed Laplacian.

Theorem 21.2 (Index of the spectral flow)**.** *The spectral flow is related to the index of the marriage Dirac operator:*

$$\mathfrak{F}_{\mathcal{A}}(\epsilon_0) - \mathfrak{F}_{\mathcal{A}}(0) = \operatorname{ind}(\mathcal{D}_{\mathrm{mar}}),$$

where $\mathcal{D}_{\mathrm{mar}}$ is the Dirac operator associated with the marriage connection.

Proof. Standard Atiyah–Patodi–Singer index theorem for spectral flow. □

Part VIII
Stratum-by-Stratum Spectral Data

22 The spectral table

Stratum δ_0	
Spectrum type	Point
Gap	> 0
Heat trace	$\mathrm{e}^{-t\lambda_0}$
Spectral determinant	λ_0
Stratum δ_1	
Spectrum type	Point + AC
Gap	resonance
Heat trace	$\sim t^{-1/2}$
Spectral determinant	–
Stratum δ_2	
Spectrum type	Point + AC
Gap	$\alpha \to 1$
Heat trace	$\sim t^{-1}$
Spectral determinant	–
Stratum δ_3	
Spectrum type	Point + AC
Gap	source
Heat trace	$\sim t^{-3/2}$
Spectral determinant	–
Stratum δ_4	
Spectrum type	Point + branch
Gap	monodromy
Heat trace	$\sim t^{-1}$
Spectral determinant	–

Stratum δ_5

Spectrum type	Point + AC
Gap	$k \to 1$
Heat trace	$\sim t^{-1}$
Spectral determinant	–

Stratum δ_6

Spectrum type	Point + AC
Gap	genus
Heat trace	$\sim t^{-9}$
Spectral determinant	–

Stratum δ_7

Spectrum type	Point + AC
Gap	memory
Heat trace	$\sim t^{-d/2}$
Spectral determinant	–

Stratum δ_8

Spectrum type	Discrete
Gap	tolerance
Heat trace	$\sim t^{-d/2}$
Spectral determinant	–

23 The spectral dictionary

Celestial quantity	**Flux quantity**	**Spectral quantity**
Orbital frequency	Kernel rate	Eigenvalue ω_n
Orbital period	Mean first passage	Heat kernel $K(t)$
Resonance	Detailed balance	Degenerate eigenvalue
Chaos	Mixing	Continuous spectrum
Action variable	Conserved flux	Bohr–Sommerfeld quantum
Stability	Ergodicity	Spectral gap

Part IX
Worked Examples

24 The Kepler problem

Example 24.1 (Kepler spectrum)**.** *The Kepler problem on $\mathcal{S}_{10} \cup \mathcal{S}_{15}$ has marriage Hamiltonian*

$$\mathcal{H}_\Sigma = -\frac{1}{2|\mathbf{p}|^2} + \frac{1}{|\mathbf{q}|}.$$

The eigenvalues of $\Delta_\mathcal{S}$ are the Rydberg energies

$$\lambda_n = -\frac{1}{2n^2}, \qquad n \in \mathbb{N}.$$

25 The harmonic oscillator

Example 25.1 (Harmonic spectrum)**.** *The harmonic oscillator on $\mathcal{S}_0$ has marriage Hamiltonian*

$$\mathcal{H}_\Sigma = \frac{1}{2}\big(|\mathbf{p}|^2 + |\mathbf{q}|^2\big).$$

The eigenvalues of $\Delta_\mathcal{S}$ are

$$\lambda_\mathbf{n} = |\mathbf{n}| + \frac{d}{2}, \qquad \mathbf{n} \in \mathbb{N}^d,$$

with degeneracy $\binom{n+d-1}{d-1}$. The marriage flow is the linear oscillator flow.

26 The elliptic stratum

Example 26.1 (Elliptic spectrum)**.** *On $\mathcal{S}_5$, the elliptic modulus k parametrizes the spectrum. The heat trace is*

$$\operatorname{Tr} \mathrm{e}^{-t\Delta_{\mathcal{S}_5}} = \frac{1}{2\pi t}\int_0^1 \frac{\mathrm{d}k}{\sqrt{k(1-k^2)}}\mathrm{e}^{-t\lambda(k)} + \cdots .$$

The spectral determinant is

$$\det \Delta_{\mathcal{S}_5} = \exp\left(-\frac{\mathrm{d}}{\mathrm{d}s}\Big|_{s=0}\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\operatorname{Tr}\mathrm{e}^{-t\Delta_{\mathcal{S}_5}}\,\mathrm{d}t\right).$$

27 The Laplace stratum

Example 27.1 (Laplace spectrum)**.** *On $\mathcal{S}_2$, the Laplace coefficient $b_s^m(\alpha)$ is the heat kernel evaluated at $t = \log(1/\alpha)$:*

$$b_s^m(\alpha) = K_{\mathcal{S}_2}(t; m, 0)\big|_{t=\log(1/\alpha)}.$$

The spectrum is discrete for $\alpha < 1$, with accumulation at $\alpha = 1$.

28 The relativistic stratum

Example 28.1 (Relativistic spectrum)**.** *On the relativistic extension of $\mathcal{S}_5$, the eigenvalues are the Schwarzschild quasi-normal modes:*

$$\lambda_n = \omega_n^2, \qquad \omega_n = \omega_R + \mathrm{i}\omega_I,$$

where ω_R is the real frequency and ω_I the decay rate. The marriage flow is the geodesic flow of the Schwarzschild metric.

29 The averaging stratum

Example 29.1 (Averaging spectrum)**.** *On $\mathfrak{H}_7$, the eigenvalues are the frequencies of the averaged Hamiltonian:*

$$\lambda_\mathbf{n} = |\mathbf{n}\cdot\omega|^2.$$

The heat kernel is

$$K_{\mathfrak{H}_7}(t;\theta,\theta') = \sum_{\mathbf{n}\in\mathbb{Z}^d} \mathrm{e}^{-t|\mathbf{n}\cdot\omega|^2}\mathrm{e}^{\mathrm{i}\mathbf{n}\cdot(\theta-\theta')}.$$

Resonances $\mathbf{n}\cdot\omega = 0$ produce zero eigenvalues.

30 The computational stratum

Example 30.1 (Computational spectrum)**.** *On $\mathfrak{H}_8$, the eigenvalues are the truncation errors:*

$$\lambda_n = \epsilon_n^2,$$

where ϵ_n is the error of the n-th approximation. The spectral flow measures the number of eigenvalues crossing zero as the tolerance is refined.

Part X
Graph and JSON Representations

31 The spectral flow graph

```
digraph SpectralFlow { rankdir=LR;

  Sigma [label="Integration_surface", shape=doublecircle];
  H [label="Interface_hypersurface_H", shape=doublecircle];
  Flow [label="Marriage_flow_phi_t"];
  Liouville [label="Liouville_equation"];
  FP [label="Fokker-Planck_equation"];
  Continuity [label="Continuity_equation"];
  Noether [label="Noether_currents"];
  Laplacian [label="Laplace-Beltrami_Delta_H"];
  Heat [label="Heat_kernel_K_H(t)"];
  Zeta [label="Spectral_zeta_zeta_H(s)"];
  Det [label="Spectral_determinant_det_Delta_H"];
  Index [label="Index_ind_D_mar"];
  WKB [label="WKB_/eikonal"];
  BS [label="Bohr-Sommerfeld"];
  EBK [label="EBK_tori"];
  Gutzwiller [label="Gutzwiller_trace"];
  Quantum [label="Quantum_marriage"];

  Sigma -> H; H -> Flow; Flow -> Liouville;
}
```

32 The spectral JSON

```
spectral_theory: {
  "interface_laplacian": {
    "operator": "Delta_H",
    "domain": "H^2(H,mu_H)",
    "spectrum": "discrete_t_continuous",
    "gap": "lambda_1_1_lambda_0"
  },
```

```
  "heat_kernel": {
    "definition": "sum_exp(-lambda_n_t)_psi_n(x)_psi_n(y)",
    "asymptotics": "4_pi_t)^(-d/2)_sum_a_k_t_k",
    "seeley_dewitt": ["Vol(H)", "R_H/6", "..."]
  },
  "spectral_zeta": {
    "definition": "sum_lambda_n^(-s)",
    "continuation": "meromorphic_on_C",
    "poles": ["d/2", "d/2-1", "...", "1", "0"]
  },
  "spectral_determinant": {
    "definition": "exp(-zeta_H'(0))",
    "topology": "chi(H)_log(4_pi)_t_...",
    "semiclassical": {
      "WKB": "A(x)_exp(i_S(x)_/hbar)",
      "eikonal": "g_{ij}_d_i_S_d_j_S_i_H",
      "bohr_sommerfeld": "oint_p_dq_i_2_pi_hbar_n",
      "EBK": "J_i_hbar_n_i_t_mu_i/4",
      "gutzwiller": "sum_over_periodic_orbits",
      "quantum_marriage": {
        "hilbert_space": "L^2(H,mu_H)",
        "hamiltonian": "-hbar^2_Delta_H_t+V_eff",
        "correspondence": "eigenfunctions_concentrate_on_Lagrangian_tori"
      }
    }
  }
}
```

Part XI
Conjectures and Research Gaps

33 Conjectures

Conjecture 33.1 (Spectral completeness)**.** *The spectrum of $\Delta_{\mathcal{S}}$ determines the marriage Hamiltonian $\mathcal{H}_\Sigma$ up to isometry. Equivalently, the interface hypersurface is spectrally rigid.*

Conjecture 33.2 (Spectral chaos)**.** *Chaos in the marriage flow is equivalent to the presence of singular continuous spectrum in $\Delta_{\mathcal{S}}$.*

Conjecture 33.3 (Goodhart and spectral gap)**.** *A Goodhart-type failure corresponds to a vanishing spectral gap:*

$$\text{Goodhart} \iff \lambda_1 - \lambda_0 = 0.$$

Conjecture 33.4 (Index and resonance)**.** *The index of the marriage Dirac operator $\mathcal{D}_{\text{mar}}$ equals the number of independent resonances on $\mathcal{S}$.*

Conjecture 33.5 (Quantum celestial mechanics)**.** *There exists a quantum celestial mechanics whose classical limit is the marriage flow and whose spectrum is the spectrum of $\Delta_{\mathcal{S}}$.*

34 Research gaps

G1. **Explicit spectrum on each stratum.** The spectrum of $\Delta_{\mathcal{S}}$ is known explicitly only on $\mathcal{S}_0$, $\mathcal{S}_1$, and $\mathcal{S}_5$. A complete computation on all strata would be valuable.

G2. **Heat kernel on $\mathcal{S}_6$.** The heat kernel on the Abelian/hyperelliptic stratum $\mathcal{S}_6$ is not explicitly known. The Seeley–DeWitt coefficients would involve the Siegel metric.

G3. **Spectral flow and algorithm complexity.** The relationship between the spectral flow $\mathfrak{F}_{\mathcal{A}}$ and the complexity of the algorithm $\mathcal{A}$ is not fully understood.

G4. **Gutzwiller trace on $\mathcal{S}$.** The Gutzwiller trace formula on $\mathcal{S}$ is formal. A rigorous treatment would require controlling the periodic orbits of the marriage flow.

G5. **Quantum–classical correspondence on $\mathcal{S}$.** The quantum marriage is defined schematically. A rigorous treatment would require constructing the Hilbert space $L^2(\mathcal{S}, \mu_{\mathcal{S}})$ and the effective potential $\mathcal{V}_{\text{eff}}$.

G6. **Index theorem on $\mathcal{S}$.** The index theorem for the marriage Dirac operator $\mathcal{D}_{\text{mar}}$ is stated but not proven. A proof would require identifying the correct K-theory class.

G7. **Spectral rigidity.** Conjecture 33.1 asserts spectral rigidity. A proof would require the marriage analogue of the Kac question.

G8. **Goodhart and spectral gap.** Conjecture 33.3 relates Goodhart failures to vanishing spectral gaps. Numerical experiments on the restricted three-body problem would test it.

G9. **Quantum celestial mechanics.** Conjecture 33.5 asserts the existence of a quantum celestial mechanics. A construction would require identifying the correct quantization of the marriage flow.

G10. **Spectral flow and resonances.** Conjecture 33.4 relates the index to resonances. A proof would require the Atiyah–Patodi–Singer index theorem on $\mathfrak{H}$.

35 Epistemic status

- **Established:** the marriage flow, its preservation of $\mathfrak{H}$, the Liouville, Fokker–Planck, and continuity equations, the Noether correspondence, the interface Laplacian, the heat kernel, the spectral zeta function, the spectral marriage theorem, the WKB/eikonal equations, the Bohr–Sommerfeld and EBK quantization, the quantum–classical correspondence.
- **Specialized:** the Arnold–Liouville locus, the KAM tori, the Gutzwiller trace formula, the spectral flow, the index theorem.
- **Derived:** the stratum-by-stratum spectral data, the spectral dictionary.
- **Equivalent:** the Hamiltonian, stochastic, and geodesic flows on $\mathfrak{H}$.
- **Potentially relevant:** the conjectures on spectral completeness, spectral chaos, Goodhart-gap, index-resonance, quantum celestial mechanics.
- **Unresolved:** the explicit spectrum on $\mathfrak{H}_3, \mathfrak{H}_4, \mathfrak{H}_6, \mathfrak{H}_7$, the heat kernel on $\mathfrak{H}_6$, the rigorous Gutzwiller trace, the index theorem.

36 Final synthesis

The dynamical and spectral theory of the interface hypersurface completes the construction begun in installments I–IV. The marriage flow is the unified dynamics of which the Hamiltonian, stochastic, and geodesic flows are projections. The Liouville, Fokker–Planck, and continuity equations are its transport laws. The Noether currents are the classical integrals of celestial mechanics. The Laplace–Beltrami operator, its heat kernel, its spectral zeta function, and its determinant are the spectral invariants of the interface. The spectral marriage theorem identifies the eigenvalues with the frequencies of the celestial-mechanics problem. The semiclassical bridge connects the classical flow to the quantum spectrum via WKB, Bohr–Sommerfeld, EBK, and Gutzwiller. The Arnold–Liouville locus and the KAM tori provide the integrable structure. The spectral flow of algorithms connects the mathematical spectrum to computational practice. The conjectures on spectral completeness, spectral chaos, Goodhart-gap, index-resonance, and quantum celestial mechanics outline the research frontier.

The completeness claim is best-evidence within the stated scope.

A The complete spectral dictionary

Celestial concept	**Flux concept**	**Spectral concept**
Orbital frequency ω	Kernel rate λ	Eigenvalue $\lambda_n = \omega_n^2$
Orbital period T	Mean first-passage time	Heat trace $\operatorname{Tr} \mathrm{e}^{-t\Delta}$
Resonance $\mathbf{n} \cdot \omega = 0$	Detailed balance	Degenerate eigenvalue
Chaos	Mixing	Singular continuous spectrum
Action variable J	Conserved flux	Bohr–Sommerfeld quantum
Stability	Ergodicity	Spectral gap
KAM torus	Invariant measure	Absolutely continuous spectrum
Homoclinic orbit	Recurrent event	Periodic orbit in Gutzwiller sum
Separatrix	Phase transition	Spectral bifurcation
Bifurcation	Critical point	Eigenvalue crossing
Attractor	Stationary measure	Ground state
Repeller	Escape measure	Continuous spectrum
Lyapunov exponent	Decay rate	Spectral gap bound
Entropy	Information rate	Spectral dimension

B Glossary

Term	Meaning
Marriage Hamiltonian $\mathcal{H}_\Sigma$	Sum of celestial, flux, and coupling Hamiltonians.
Marriage vector field $X_{\mathcal{H}_\Sigma}$	Hamiltonian vector field of $\mathcal{H}_\Sigma$.
Marriage flow ϕ_t	One-parameter group generated by $X_{\mathcal{H}_\Sigma}$.
Liouville equation	$\partial_t \rho + \{\rho, \mathcal{H}_\Sigma\} = 0$.
Fokker–Planck operator	Generator of the marriage diffusion.
Marriage current J_{mar}	$\rho X_{\mathcal{H}_\Sigma} + D\nabla\rho$.
Marriage charge Q_{mar}	Integral of $\mathcal{H}_\theta$ on $\mathcal{S}$.
Interface Laplacian $\Delta_{\mathfrak{H}}$	Laplace–Beltrami operator of $g_{\mathfrak{H}}$.
Heat kernel $K_{\mathfrak{H}}(t; x, y)$	Fundamental solution of the heat equation.
Spectral zeta $\zeta_{\mathfrak{H}}(s)$	$\operatorname{Tr} \Delta_{\mathfrak{H}}^{-s}$.
Spectral determinant	$\exp(-\zeta'_{\mathfrak{H}}(0))$.
Eikonal equation	$g_{\mathfrak{H}}^{ij} \partial_i S \, \partial_j S = \mathcal{H}_\Sigma$.
Bohr–Sommerfeld	$\oint p \, \mathrm{d}q = 2\pi\hbar n$.
EBK torus	Lagrangian torus with $J_i = \hbar(n_i + \mu_i/4)$.
Gutzwiller trace	Semiclassical density of states.
Quantum marriage	$L^2(\mathfrak{H}, \mu_{\mathfrak{H}})$ with Hamiltonian $-\hbar^2 \Delta_{\mathfrak{H}} + \mathcal{V}_{\mathrm{eff}}$.
Arnold–Liouville locus	Integrable locus of $\mathfrak{H}$.
KAM torus	Invariant Lagrangian torus under perturbation.
Spectral flow	Number of eigenvalues crossing zero.
Index $\operatorname{ind} \mathcal{D}_{\mathrm{mar}}$	Index of the marriage Dirac operator.

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